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Jeffrey L Crabtree, Joseph K Wells

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Trends Affecting Entry Level Occupational Therapy Education in the United States of America and Their Probable Global Impact

Author : Joseph K. Wells* OTD, OTR/L, DPMIR, **Co-Author :** Jeffrey L. Crabtree**, OTD, MS, OT, FAOTA

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*Co-owner/ Governing Executive,
Sohm America Integrations Group

** Associate Professor, Department
of Occupational Therapy, School of
Health and Rehabilitation Sciences,
Indiana University—Purdue
University Indianapolis,
Indianapolis, Indiana, USA

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Correspondence to :
Joseph K. Wells, OTD, OTR/L,
DPMIR*
Co-owner/ Governing Executive,
Sohm America Integrations Group
Adjunct Faculty, Owens Community
College, 1440 S. Byrne Rd.,
Toledo, OH 43614, USA

E- Mail :
jwells@theSAlgroup.org

Abstract

Academic trends and practice requirements have caused many healthcare professions to mandate a doctoral entry (also called the first-professional doctorate). The American Occupational Therapy Association (AOTA) currently maintains a two-point entry to practice at the Master's and Doctoral levels in the United States of America (US). This paper evaluates the probable impact of AOTA's current policy on the profession both within the US and internationally. The authors also suggest a pathway to facilitate development and exchange of a global workforce.

Introduction

Professional entry to occupational therapy practice in the US requires either a master's degree or clinical doctorate. The master's degree is usually credentialed as *Master of Occupational Therapy*, abbreviated *MOT*, or *Master of Science in Occupational Therapy*, abbreviated *MS* or *MSOT*. The doctoral degree is usually titled as *Doctor of Occupational Therapy* or *Occupational Therapy Doctorate*, abbreviated as *OTD*. These degrees are also offered as post-professional or advanced degrees for those already in practice.

Practitioners now are required to be more prepared than ever to address healthcare complexities caused by increased knowledge base, longevity and diversity of population, and technological advances (Royeen and Lavin 2007).¹ The services that a profession may render are commonly defined by its scope of practice. Today with the existence of several health disciplines, it is impractical to expect a unique scope of practice limited to any one profession. In fact, changes in the scope of practice are axiomatic and overlapping practice areas are not only inevitable but necessary to perform the duties required of a healthcare provider,² especially considering the shortages of all sorts of providers globally. The educational system of a profession is determined by its scope, role and function it wishes to play (Fidler 1966).³ Conversely, education and training are determinants of its scope of practice. The other factors influencing a profession's scope are its established history of the practice scope within the profession, supporting evidence to demonstrate historical practice and educational/ training preparation to safely deliver the services it claims within its scope and appropriate regulatory environment.² Occupational therapy must therefore be able to continually articulate why it provides the service it does, and in the face of competition, is the best suited to do so.

To better understand the trends affecting occupational therapy's education and practice, it may be helpful to get a perspective on the influencing factors by looking into its past.

The basis of occupational therapy is rooted in ancient times. Sidney Licht (in Quiroga, 1995) traces the earliest evidence to one hundred years before Christ. It was around this time when the Greek physician Asclepiades initiated treatment of patients with mental illness using activity based modalities. More recently, in 18th century Europe, Phillipe Pinel, Johann Christian Reil, and William Tuke advocated for moral treatment approaches versus established treatment regimens that included restraints, torture and punishment in the treatment of persons with mental illness. This reform movement spread momentarily to the American side in the early and mid 19th century.⁴ However, it wasn't until the turn of the 20th century that modern day occupational therapy was formalized with the revival of the reform movement (Kearney 2005).⁵ In 1917, the National Society for the Promotion of Occupational Therapy was formed that was later renamed the American Occupational Therapy Association (AOTA).⁵ Later that year, reconstruction aides in occupational therapy were placed in the US military.

The first set of educational standards was proposed by the US Surgeon General in 1918 requiring a ten- week (McDaniel 1968 in Kearney 2004) program emphasizing leisure and recreational

crafts and was strongly grounded in the moral treatment model. However, the need to also have medical knowledge was identified quickly and the standards were revised later that year. This then began the profession's educational contract with the medical model as well.⁵ It is since then that occupational therapy curriculum and practice have been influenced by two paradigms, one based on the moral treatment model that emphasized the rights of humans and humanistic approach to treatment and the other on the medical model that emphasized healing the human body by understanding and treating the various components that comprise it.⁵ Historically, these paradigms have been in conflict with the medical model being overtly physician-centered.^{5,7} This conflict has caused much confusion and debate within the profession that has swung between both these paradigms in its efforts to define its educational and practice requirements over the years.^{5,6}

The advancements in occupational therapy education and practice are integrally related to medicine. Scientifically based medical education with a commitment to research was instituted by what is commonly known as the Flexner Report in 1910. This report revolutionized medical education in the US requiring prescribed standards. As a result, by early 1920s, medical schools were no longer allowed in proprietary institutions and were required to be university based. Similarly, many occupational therapy schools, by the late 1920s, were affiliated to colleges and universities. It was in 1923 when AOTA published the first educational standards for non-emergency war courses.⁵ From 1931 to 1994, AOTA entered into a collaborative relationship with the American Medical Association (AMA) to accredit its educational programs.⁵ This led to the profession being closely linked to the medical model. Educational standards prescribed during this period were in the form of documents known as the *Essentials of an Acceptable School of Occupational Therapy* ("Essentials") that were updated periodically. The last Essentials was published in 1991; a revision in 1995 simply announced the break from AMA programs as by then AOTA's Academic Council for Occupational Therapy Education was recognized by the US Department of Education and Commission on Recognition of Post Secondary Education as an independent accrediting agency for occupational therapy programs.⁵

Method

Using the keywords, we searched the internet periodically between September 2009 and November 2011 on Google, Google Scholar and PubMed (*Table- 1*). Due to the high volume of search results on Google and Google Scholar, we visually scanned articles from the first ten pages of these searches. The

PubMed search results were visually scanned for relevance. Articles found to be potentially relevant were further selected and full-text versions were obtained where needed. Online searches included focused search on websites of US national associations of professions that have mandated or are transitioning to doctoral entry in the near future. Online searches were also conducted on focused queries such as level of education and outcomes in healthcare, impact of higher qualification requirements for professional entry in healthcare on minorities and immigration, employment and level of education, social issues pertaining to higher qualification requirements for healthcare providers, historical bases of occupational therapy education, history of medicine, scopes of practice and competition amongst health professions, et cetera. We also manually searched and selected articles from journals or trade publications such as the American Journal of Occupational Therapy, OT Practice, Advance for Occupational Therapy Practitioners, Today in OT, et cetera. A few books were also included in the selection of the literature we reviewed. We discussed findings with other healthcare professionals to arrive at the final list of sources.

Results

The results of the last online search in November of 2011 are posted in *Table-1*. Based on the review of relevant literature and qualitative processes used as described under "Material and Method", it is apparent that multiple healthcare professions in the US are soon transitioning or have transitioned to a doctoral entry to practice. Occupational therapy currently is not one of them. The benefits or harm from a doctoral entry mandate have not been adequately tested quantitatively mostly due to the novelty of the program itself. However, they may be discussed based on historical factors, societal and marketing perspectives, and developmental aspects of a profession.

Discussion

Over the years occupational therapy educational curricula underwent many changes. In 1999, AOTA's Representative Assembly (RA) passed a directive known as Resolution- J that mandated post-baccalaureate entry at the professional level.⁸

Unlike most British Commonwealth countries, the US has required a doctoral entry in medicine, osteopathy, dentistry, optometry, clinical psychology, podiatry, veterinary and chiropractic for many years now. Other professions have recently transitioned or are soon transitioning to doctoral entry. Pharmacy leads this pack having transitioned to doctoral entry in 2004-

Table 1.
Number of articles that resulted from online searches

Keywords	Google	Google Scholar	Pub Med
Clinical doctorate	8,690,000	42,100	183
Clinical doctorate AND occupational therapy	2,550,000	20,500	6
Clinical doctorate AND healthcare	4,350,000	27,200	57
Professional doctorate	24,600,000	101,200	129
Professional entry and clinical doctorate	4,420,000	26,100	6

2005,⁹ with Audiology following the trend by upgrading its entry to a doctoral level in 2012.¹⁰ Also, all entry programs for advance practice nurses¹¹ and physical therapists¹² will offer doctoral degrees after 2015. Witnessing the shift to doctoral entry by these professions, the question is inevitable- *should we or shouldn't we?*

AOTA's RA Resolution-J supported by a study conducted under RA Charge-111 by Coppard et al. (2009) currently maintains that occupational therapists beginning 2007 can enter practice in the US through a two-point entry, that is, at either the master's or the doctoral level.¹³ Many currently believe that mandating a doctoral entry in occupational therapy should be avoided as it merely follows trends based on the medical model underlining the profession's continuing "identity crisis" (Gilfoyle 1984)¹⁴ trying to find a balance between the paradigms of moral treatment and the medical model which it views as conflicting with each other.⁵ Many also feel that the move to a single-point doctoral entry will mean "simply following physical therapy" (Glomstad 2008, p.12).¹⁵ Holders of these beliefs probably feel that the medical model makes the profession reliant on medicine for the development of its own concepts supporting its practice.⁶

We believe that entry to occupational therapy practice should be singularly at the doctoral level. While this debate certainly needs to be happening urgently in the US as it has direct consequences here, it is also of global interest since health policies and practice are often based on international referencing and networking of professionals, and organizations such as the World Health Organization (WHO). We have focused our discussions here to factors that have more global relevance versus those that are strictly limited to the US.

Interest in the Educational Levels. We found no studies that compared student interest between different healthcare professions, or for those specifically applying to occupational therapy schools, to help us gain insight to whether the degree itself or duration of an educational program plays a role in its selection. But assuming that the interest in different professions is the same, there is growing interest in occupational therapy at both the master's and doctoral levels. There was a 25% increase in the entry doctorate and 7% in the entry master's enrollments for 2009-10 as compared to 2008-09 (AOTA 2009, 2010).^{16, 17} This indicates a growing interest in the doctoral entry despite the unequal access with just four out of one hundred and forty six schools offering entry doctorates in the US.¹⁷ This is also noteworthy since entry doctorates were disadvantaged until recently due to an AOTA moratorium on accrediting any new entry doctoral programs between 2005 and 2009. Also according to the US Census Bureau, from 1990 through 2006, more resident aliens graduated from doctoral programs (23.2%) in the US than from master's programs (10.9%).¹⁸ Doctoral programs in the US, unlike those that follow the British system, usually allow admission after a baccalaureate degree. This feature probably is favorable to many who wish to obtain a doctoral degree to begin with.

Educational Outcomes and Market Forces. There is currently a paucity of both theoretical and empirical data on outcomes

pertaining to the two entries in occupational therapy practice owing to the fact that entry-doctorates are fairly new.^{13, 19} However, Smith (2007) indicated that the doctorate is perceived to be favorable toward obtaining higher salaries, career advancement and professional competence.²⁰ A meta-analysis favorably linked the level of education to better communication skills, knowledge, problem-solving and professional skills in registered nurses (RNs) trained at the bachelor's versus associate's or diploma levels.²¹ Studies also found that hospitals with higher percentage of RNs trained at the bachelor or higher level versus the associate or diploma level had lower mortality rate and failure-to-rescue rates with surgical patients.^{22, 23} That is, a higher level of education of practitioners may help realize better patient outcomes. From a market interest perspective, Skipper & Lewis (2006) found that employers were more interested to hire registered dietitians with clinical doctorates than without.²⁴ And, even AOTA (2009) indicated that 80% of entry-doctoral graduates compared to 33% of master's found employment by the second week of graduation. These data certainly support the doctoral entry in occupational therapy.²⁶

In order to discuss the market forces affecting the profession, we must also discuss competing factors and trends in other professions that are historically similar in nature. Occupational therapists and physical therapists share common roots as "reconstruction aides" in the US military that led to their establishment and growth during and after World War I.⁶ Although occupational therapy had a longer and more complex history (Gritzer & Arluke 1985, p. 54),⁶ physical therapy currently is spread in more countries with 106 member organizations²⁵ compared to occupational therapy with 69 member organizations.²⁶ Also, in the US, while occupational therapy had a more autonomous status prior to World War I, currently occupational therapy has direct access, the ability to see patients without a referral from a physician, in lesser states than physical therapy. Occupational therapy has direct access in 34 states²⁷ compared to 46 for physical therapy.²⁸

Although, the mandate on doctoral entry in physical therapy is not effective until December 31, 2015, currently 203 (96%) of physical therapy schools offer doctoral entry.²⁹ By contrast, only 4 (3%) occupational therapy schools offer entry level doctorates.¹⁷ According to the Association of Schools of Allied Health Professions (ASAHP), requirement for doctorally trained clinicians carries with it the understanding that the practitioners are more highly qualified to provide health care by virtue of their additional years of training (2009, p.2).³⁰ From a market perspective, if occupational therapy does not move in favor of a doctoral-only entry fast enough, its practitioners may draw lowered public confidence as the providers of choice for services with overlapping activities in its scope. It may also face greater challenges to its scope of practice.

Educational Standards. The last few decades witnessed some of the greatest advancements in medicine. While many diseases do not pose the same level of fatality as they did before, challenges to healthcare certainly exist and in no lower intensity, for example, the emergence of HIV/AIDs as a major disease

since the 1980s, and the recognition of the infectious disease caused by the Ebola virus for the first time in 1976.³¹ Technological advances of the last two decades alone, such as communications, imaging and computer technology, genetics and biotechnology, robotics et cetera, have caused a major expansion of knowledge base requiring more entry-level training in the health professions.³⁰

Pharmacy, physical therapy, audiology and advance practice nursing, all cite the advances in healthcare as a reason for upgrading their entry education to a doctoral degree. In fact, Coppard (in Bollag, 2007) advocated strongly for the entry doctorate in occupational therapy based on the growth of the knowledge base and practice demands, and helped create the first program instituted at Creighton University in 1999.³²

Higher training at the doctoral level is viewed to facilitate clinical skills and patient care.³⁰ Entry doctorates in physical therapy or occupational therapy are nine to twelve months more in duration than the current master's. One of the rationales for mandating doctoral entry by the American Physical Therapy Association was based on the fact that the existing master's programs had enough or close enough credits to warrant a clinical doctorate.³³ One must therefore ask if occupational therapy students in the US are being deprived of a degree they are probably more suited for.

A profession is mostly recognized by a body of knowledge required by its practicing members.³ AOTA's resolve to maintain a two-point entry in occupational therapy to groom professionals with exactly the same scope of practice is ambiguous, and probably gives the impression that the knowledge base of occupational therapy is inferior compared to other professions transitioning or transitioned to a doctoral entry.

Social Issues and Immigration. AOTA aspires to be a globally connected and diverse workforce.³⁴ Coppard et al. (2009, p.13) opposed the singular entry in occupational therapy at the doctoral level feeling that it "would not allow [foreign trained therapists] to work and share their expertise in this country".¹³ The educational preparation for entry to practice and employment requirements vary throughout the world.^{35,36} The World Federation of Occupational Therapists (WFOT) recognizes and supports the various levels of entry that are aligned with the educational preparation of domestic students, the academic credentials of similar professions, expectations for health professionals and salary levels available for qualified health professionals in specific context and region (p.1).³⁵ One of the three considerations for countries proposing higher entry-level education as identified by WFOT are related to the barriers it may cause to graduates from other member countries seeking to practice there.³⁵

Beginning July 1, 2013, the National Board of Certification in Occupational Therapy requires foreign trained occupational therapists seeking registration to practice in the US to have master's degree.³⁷ The argument that a doctoral entry will pose severe barriers to foreign graduates as compared to a master's entry is exaggerated. The disparity in professional entry is not new globally. For example, entry to practice medicine in the

US is at the doctoral level while in most British Commonwealth countries it's with a bachelor's degree that usually takes five to six years after a high school diploma.³⁸ AMA has worked with this discrepancy in educational preparation and credentials for many years. It allows a pathway to certify foreign graduates through the Educational Commission for Foreign Medical Graduates after one to four years of residency based upon the specialization selected.³⁹ Upon successful completion of the residency and licensure requirements, foreign medical graduates attain the same credentials of *Medical Doctor* or *Doctor of Medicine*, abbreviated *MD*, similar to a US trained physician although their actual academic degree is different. Also, success to attract foreign talent in medicine to the US is well evident. In 1998, approximately 23% of all physicians, 24% of residents and 37% of interns were international medical graduates.⁴⁰

We propose that similar to AMA, AOTA should develop a pathway that allows transferability of foreign graduates who meet the Minimum Standards prescribed by WFOT irrespective of the educational degree earned. The process should involve preparing the graduates adequately through programs similar to medical residencies to equip them with the needs to practice competently in the US. Currently the responsibility and burden to prepare foreign graduates to enter practice in a country is mainly borne by the immigrant and his/her employer. We recommend that before unrestricted practice is commenced, a prescribed university or equivalent workplace based program be required whose content and duration shall depend on the practice demands and cultural requirements. This would best prepare the practitioners and facilitate optimal patient safety.

Also, with an ever shrinking international occupational therapy community aided by the internet, students and practitioners have access to global expertise more than ever, for example, the concepts of client-centered practice and the Canadian Occupational Performance Measure are available through a number of international journals.^{41,42,43,44,45,46,47,48,49,50,51} That is, a doctoral entry mandate has no effect on the flow of expertise globally in this respect.

Countries must train practitioners in accordance to its prevailing health needs, priorities, and social and community factors. The Minimum Standards published by the WFOT while supporting transferability of skills and fostering of international cooperation also supports proper educational content and processes to meet the local health needs.³⁵ Adhering to recommendations by WFOT, AOTA must ensure that entry-level education meet the demands of the American society and prepares practitioners for the roles expected of them. In doing so, it must also consider the educational credentials of historically similar professions in the US.³⁵ Modeling after the US, Pakistan and Australia have recently started offering entry-level doctorates in physical therapy. Similarly AOTA's actions will affect the status of the profession globally. AOTA's resolve to maintain a master's entry as an option just because no doctoral entry programs are available internationally¹³ holds the development of OT hostage in the US, and by extension internationally.

Conclusion

The 20th century witnessed the medical profession's overhaul of its educational system and commitment to scientific practice which was further strengthened through the evidence-based medicine movement of the 1990s. Evidence-based medicine not only emphasizes scientific practice but also the assimilation of patient values and interests (Sackett et al. 1996).⁵² Medicine has certainly become more conscientious toward patient-centered practice in the 21st century.^{7,53} As reflected by the WHO's International Classification of Functioning, Disability and Health published in 2001, medicine recognizes the role of the environment on health and wellness.⁵ The medical model and moral treatment paradigms can no longer be viewed as conflicting to each other but rather complementing. Occupational therapy's aspiration to be science-driven and evidence-based³⁴ requires it to be a partner in the medical model.⁵ And, its focus on the occupational needs with patient-centered approach allows it to be rooted in the moral treatment model.⁵ We believe that there is ample evidence in the literature as reported here for AOTA to recognize the educational trends in healthcare and quickly learn to thrive in it. It is important to make a candid introspection into the history of the profession that mark our market mistakes such as opposing licensure, inability to clearly define our services, and inability to influence Medicare legislation in the 1960s and early 1970s in the US. Noting these mistakes along with the inability to define our product and public ambiguity related to it, Gritzer and Arluke (1985) warned that unless the profession is able to correct their market strategy, it "will remain subordinated to other groups in the division of labor" (p.145).⁶ We believe that this statement is still relevant to the profession in the US as it is yet to achieve licensure in all the states, has fewer states that allow direct access thus representing its subservience to medicine, and also is yet to become an initiating service in the home under Medicare unlike physical therapy, speech-language pathology and nursing.

We urge that AOTA recognize that a two-point entry only maintains an unwanted status-quo and confusion. It serves to appease both the detractors and proponents of the single-point entry- doctorate but fails to articulate what exactly is the core knowledge base and skills set required to enter practice. Pharmacy, audiology, physical therapy and advance practice nursing quote needs of the society and advances in their knowledge base as major contributory factors toward doctoral entry. Occupational therapy exists in a dynamic healthcare environment that demands practitioners be autonomous, culturally competent, with expanded geriatric and neuroscience knowledge to address the complexities brought by greater longevity.¹ They also must have superior inter-professional skills to collaborate with other healthcare team members, and have a deep understanding of health informatics that go beyond the basics.¹ We assert that the international community is best served by preparing occupational therapists with the highest standards of education and training with credentials comparable to their peers. And, concur with Royeen and Lavin (2007, p.611), "Occupational therapy must move to a doctoral level of entry if we are to remain a viable profession".¹

References

1. Royeen CB, Lavin MA. A contextual and logical analysis of the clinical doctorate for health practitioners: dilemma, delusion, or de facto? *J Allied Health*. 2007; 36, 101-6.
2. National Council of State Boards of Nursing. *Changes in healthcare professions' scope of practice* [Internet]. Chicago: Author; 2009 Oct [cited 2011 Dec 4]. Available from: <https://www.ncsbn.org/ScopeofPractice.pdf>
3. Fidler GS. *Learning as a growth process: a conceptual framework for professional education: a conceptual framework for professional education*. *Amer J Occ Ther*. 1966; 20:1-8.
4. Quiroga VAM. *Occupational therapy: the first 30 years- 1900 to 1930*. Bethesda, MD: American Occupational Therapy Association. 1995: p.19-88.
5. Kearney P. *The influence of competing paradigms on occupational Therapy education: a brief history* [Internet]. Unknown: Author; 2005 Oct [cited 2011 Dec 4]. Available from: <http://www.newfoundations.com/History/OccTher.html>
6. Gritzer G, Arluke A. *The making of rehabilitation*. Berkeley: University of California Press. 1985; p.52-145.
7. Laine C, Davidoff F. Patient-centered medicine: a professional evolution. *JAMA*. 1996; 275:152-6
8. Accreditation Council for Occupational Therapy Education. *ACOTE Motion and Resolution J* [Minutes at the meeting of the Accreditation Council for Occupational Therapy Education]. Bethesda, MD: Author. 1999, Aug.
9. Accreditation Council for Pharmacy Education (2006). *Accreditation standards and guidelines for the professional program in pharmacy education leading to the doctor of pharmacy degree* [Internet]. Chicago: Author, 2006 Feb [cited 2011 Dec 4]. Available from: http://www.acpe-accredit.org/pdf/ACPE_Revised_PharmD_Standards_Adopted_Jan152006.pdf
10. American Speech-Language-Hearing Association. *2007 Standards for the certificate of clinical competence in audiology* [Internet]. Washington, DC: Author, 2009 Mar [cited 2010 Sep 4]. Available from http://www.asha.org/about/membership-certification/certification/aud_standards_new.htm
11. American Association of Colleges of Nursing. *Frequently asked questions: doctor of nursing practice (DNP) programs* [Internet]. Washington, DC: Author, 2009 Oct [cited 2011 Dec 4]. Available from <http://www.aacn.nche.edu/DNP/DNPFQA.htm>
12. American Physical Therapy Association. *Physical therapist (P.T.) education overview* [Internet]. Alexandria, VA: Author, 2011 Mar 10 [cited 2011 Oct 14]. Available from: <http://www.apta.org/PTEducation/Overview/>
13. Coppard B, Berthelette M, Gaffney D, Muir S, Rietz SM, Slater DY. Why continue 2 points of entry education for occupational therapists? *OT Prac*. 2009; 14(4), 10-4.
14. Gilfoyle EM. Transformation of a profession. *Amer J Occ Ther*. 1984; 38(9), 575-84.
15. Glomstad J. Examining the evidence: supporting both master's and doctoral level entry into the profession. *Adv OT Prac* [Internet]. 2008 Dec 8, 2008 [cited 2011 Dec 4]; 24 (25): 12. Available from: <http://occupational-therapy.advanceweb.com/Article/Examining-the-Evidence-5.aspx>
16. American Occupational Therapy Association. *Academic programs annual data report: academic year 2008-2009* [Internet]. Bethesda, MD: Author. 2009 [cited 2011 Oct 13]. Available from: <http://www.aota.org/Educate/Accredit/47682/44045.aspx?FT=.pdf>
17. American Occupational Therapy Association. *Academic programs annual data report: academic year 2009-2010* [Internet]. Bethesda, MD: Author. 2009 [cited 2011 Oct 13]. Available from: <http://www.aota.org/Educate/Accredit/47682/46227.aspx?FT=.pdf>
18. US Census Bureau. *The 2010 Statistical Abstract: the national handbook* [Internet]. Washington, DC: Author. Not dated [cited 2011 Dec 4].

- Available from <http://www.census.gov/compendia/statab/2010edition.html>.
19. Ellis L. Academics' perceptions of the professional or clinical doctorate: findings of a national survey. *J Clin Nurs*. 2007; 16, 2272-9.
 20. Smith DL. Perceptions by practicing occupational therapists of the clinical doctorate in occupational therapy. *J Allied Health*. 2007; 36, 137-40.
 21. Johnson JH. Differences in the performances of baccalaureate, associate degree, and diploma nurses: a meta-analysis. *Res Nurs Health*. 1988; 11(3), 183-97.
 22. Aiken LH, Clarke SP, Cheung RB, Sloane DM, Silber, JH. Educational levels of hospital nurses and surgical patient mortality. *JAMA*. 2003; 290(12), 1617-23.
 23. Long KA, Bernier S. RN education: a matter of degrees. *Nurs*. 2004; 34(3), 48-51.
 24. Skipper A, Lewis, NM. Clinical registered dietitians, employers, and educators are interested in advanced practice education and professional doctorate degrees in clinical nutrition. *J Amer Diet Assoc*. 2006; 106, 2062-6.
 25. World Confederation of Physical Therapy. About WCPT [Internet]. London: Author. 2011 [updated 2011 Jun 23; cited 2011 Dec 4]. Available from: <http://www.wcpt.org/about>
 26. World Federation of Occupational Therap. Member organizations of WFOT [Internet]. Forrestfield: Author. 2011 [cited 2011 Dec 4]. Available from: <http://www.wfot.org/Membership/MemberOrganisationsofWFOT.aspx>
 27. AOTA State Affairs Group. Occupational therapy- referral requirements [Internet]. Bethesda, MD: American Occupational Therapy Association. 2008 Jul [cited 2011 Dec 4]. Available from: <http://www.aota.org/Practitioners/Licensure/StateRegs/Referral/36361.aspx?FT=.pdf>
 28. American Physical Therapy Association. FAQ: direct access at the state level [Internet]. Alexandria, VA: Author. 2011 [updated 2011 Nov 11; cited 2011 Dec 4]. Available from: <http://www.apta.org/StateIssues/DirectAccess/FAQs/>
 29. American Physical Therapy Association. Physical therapist (P.T.) education overview [Internet]. Alexandria, VA: Author. 2011 Mar 10 [cited 2011 Dec 4]. Available from: <http://www.apta.org/PTEducation/Overview/>
 30. Association of Schools of Allied Health Professions. Potential impacts of entry level clinical doctorate degrees in the health professions [Internet]. Washington, DC: Author. 2009 [cited 2011 Dec 4]. Available from: <http://www.asahp.org/pdf/Clinical%20Doctorate%20Impacts%20Paper.pdf>
 31. Chris T. History of medicine. UK: History Learning Site [Internet]. c2000-2011 [cited 2011 Dec 4]. Available from: http://www.historylearningsite.co.uk/medicine_and_world_war_two.htm
 32. Bollag B. Credential creep. *Chr Higher Educ* [Internet]. 2007 Jun 22 [cited 2011 Dec 4]. Available from: <http://chronicle.com/article/Credential-Creep/25476>
 33. American Physical Therapy Association. Education programs: doctor of physical therapy (DPT) program-frequently asked questions [Internet]. Alexandria, VA: Author. Not dated [cited 2010 Feb 15]. Available from: <http://www.apta.org/AM/Template.cfm?Section=Clinical&CONTENTID=69547&TEMPLATE=/CM/ContentDisplay.cfm#BM6>
 34. American Occupational Therapy Association. AOTA's centennial vision [Internet]. Bethesda, MD: American Occupational Therapy Association. 2006 Jan 20 [cited 2010 Dec 27]. Available from <http://www.aota.org/News/Centennial/Background/36516.aspx?FT=.pdf>
 35. World Federation of Occupational Therapy. Position statement: occupational therapy entry-level qualifications [Internet]. Forrestfield: Author. 2008 [cited 2011 Dec 4]. Available from: <http://www.wfot.org/ResourceCentre.aspx>
 36. World Federation of Occupational Therapy. Requirements for the employment of occupational therapists in member countries of the World Federation of Occupational Therapy. Forrestfield: Author. 1986 [revised 2007; cited 2011 Dec 4]. Available from: <http://www.wfot.org/ResourceCentre.aspx>
 37. National Board of Certification in Occupational Therapy. International-services [Internet]. Gaithersburg, MD: Author. c2009 [cited 2011 Dec 4]. Available from: http://www.nbcot.org/index.php?option=com_content&task=view&id=281&Itemid=10
 38. Educational Commission for Foreign Medical Graduates. 2010 information booklet: reference guide for medical education credentials [Internet]. Philadelphia, PA: Author. C2001-2011 [updated 2011 Sep 15; cited 2011 Dec 4]. Available from: <http://www.ecfm.org/creds/refgde.html>
 39. Educational Commission for Foreign Medical Graduates. 2010 information booklet: reference ECFMG certification [Internet]. Philadelphia, PA: Author. C2001-2011 [updated 2011 Sep 15; cited 2011 Dec 4]. Available from: <http://www.ecfm.org/2010ib/ibcert.html>
 40. Shepherd S. Dr. IMG: diverse international graduates attempt to gain recognition in us health care. *New Phys* [Internet]. 2002; 51(7). Available from: <http://www.amsa.org/AMSA/Homepage/Publications/TheNewPhysician/2002/tnp197.aspx>
 41. Canadian Association of Occupational Therapists. Enabling occupation: An occupational therapy perspective. Toronto, Ontario: CAOT Publications. 1997.
 42. Farnworth L. Sylvia Docker Lecture: Time use, tempo and temporality: Occupational therapy's core business or someone else's business? *Aus Occ Ther J*. 2003; 50(3), 116-26.
 43. Jansa J, Sicherl Z, Angleitner K, Law M. The use of Canadian Occupational Performance Measure (COPM) in clients with an acute stroke. *World Fed Occ Ther Bulletin*. 2004; 50, 18-23.
 44. Hammell, KW. Application of learning theory in spinal cord injury rehabilitation: Client-centered occupational therapy. *Scand J Occ Ther*. 1995; 2(1), 34-9.
 45. Law M, Baptiste S, Carswell A, McColl MA, Polatajko H, Pollock N. (1998). The Canadian occupational performance measure. 3rd. ed. Toronto, Ontario: CAOT Publications. 1998.
 46. Law M, Baptiste S, Mills J. Client-centered practice: what does it mean and does it make a difference? *Can J Occ Ther*. 1995; 62, 250-7.
 47. Morgan SB, Kelkar RS, Vyas OA. Client-centered occupational therapy for acute stroke patients. *Indian J Occ Ther*. 2002; 34(1), 7-12.
 48. Rigby R, Ryan S, From W, Walczak E, Jutai J. A client-centered approach to developing assistive technology with children. *Occup Ther International*. 1996; 3(1), 67-79.
 49. Unsworth CA. Clinical reasoning: how do pragmatic reasoning, worldview and client-centredness fit? *British J Occup Ther*. 2004; 67(1), 10-9.
 50. Warren A. An evaluation of the Canadian Model of Occupational Performance and the Canadian Occupational Therapy Measure in mental health practice. *British J Occup Ther*. 2004; 65(11), 515-22.
 51. Wressle E, Samuelsson K, Henriksson C. Responsiveness of the Swedish version of the Canadian Occupational Performance Measure. *Scan J Occup Ther*. 1999; 6(2), 84-9.
 52. Sackett DL, Rosenberg WMC, Gray JAM, Haynes RB, Richardson WS: Evidence based medicine: what it is and what it isn't. *BMJ* 1996; 312:71-2.
 53. Ullman K. Patient-centered medicine can benefit primary care practices. *DOC News* [Internet]. Alexandria, VA: American Diabetes Association. 2006 Oct; 3 (10), 6-7 [cited 2011 Dec 4]. Available from: <http://docnews.diabetesjournals.org/content/3/10/6.full>